Penetration – The penetration test is used to measure the hardness of bitumens. A lower penetration number indicates greater hardness.

In the test, a container of bitumen is kept at 25° C (77° F). A steel needle is applied to the surface of the bitumen for five seconds under a load of 100 grams. The distance the needle penetrates, rated in tenths of a millimeter (dmm), is the penetration measurement.

Specifications for penetration-graded bitumens typically state the penetration range. On the basis of the penetration test, bitumens are classified into five standard grades from hardest to softest: 40-50, 60-70, 85-100, 120-150, and 200-300 dmm.

Viscosity – Viscosity is typically calculated using the time required for the bitumen to flow between two successive marks. The viscosity of a bitumen is graded in hundreds of poises (grams per centimeter per second) at 60° C (140° F).



User Tip: *From the standpoint of the paving crew, knowing the hardness and viscosity of the bitumen helps crewmembers understand how the mixture will flow under and support the screed. The harder or more viscous the bitumen, the stiffer the mixture will be. A stiffer mix tends to support the screed more easily and the crew may have to adjust accordingly. Those screed adjustments will be discussed in Unit 3, "Fundamentals of Paving."*

In the United States during the 1990s, the Strategic Highway Research Program introduced the performance-grade system to replace the penetration and viscosity grading systems. The performance-grade system is used to assess and designate the engineering properties of bitumens at temperatures that are representative of climatic conditions in which the bitumens will be used.

Performance grade is a two-number system where the first number represents the maximum pavement design temperature (C°), while the second number represents the minimum likely pavement design temperature (C°) that can be used without failure.

For example, performance grade 64-28 signifies maximum pavement temperature at 64° C (150° F) and minimum pavement temperature at -28° C (-20° F). A bitumen classed as 64-28 provides enough stiffness to prevent permanent deformation or rutting temperatures as high as 64° C (150° F). Yet, a 64-28 bitumen remains flexible enough to resist cracking at temperatures as low as -28° C (-20° F).

Most paving bitumens are classified as straight-run bitumens. They are usually produced from the residue of atmospheric distillation of petroleum crude oil. Additional classes include oxidized bitumens, cutback bitumens, and bitumen emulsions. These last three classes are not typically used for the production of high-temperature bituminous paving material.

Another class, known as modified bitumens, contains quantities of special additives such as polymers, crumb rubber, latex, sulfur, polyphosphoric acid and/or other products that modify their properties to meet specific engineering requirements. Modifiers typically represent 3 to 15 percent of the bitumen by weight.

User Tip: *Modifiers such as polymer and latex tend to make mixes stiffer. They are often produced at higher than normal temperatures to help the mixes flow under the screed and as an aid to the compaction process. Some modified mixes are very sticky. The frequency of cleaning and maintenance must be increased to avoid build-up of mix on the paver's feeder system and screed components.*

[AGGREGATES]

While bitumens serve as the binding agent in the bituminous mixture, aggregates serve as the load-bearing component of the bituminous layer. Various types of aggregates are used in bituminous paving mixtures. They are classified as sand, gravel, crushed stone, slag and mineral filler.



Sand – Sand is the fine granular material resulting from the natural disintegration of rock or the crushing of rock. There are many different types of sand, both naturally occurring and artificial. Each type of sand has its own characteristics. Sands often are blended to achieve the desired gradation for a particular mix formula.

Sands are classed by size range, often being between 0.05 and 2 mm (0.002 and 0.08") with 2 mm (0.08") being the opening of a #10 mesh sieve and 0.05 mm (0.002") being the opening of a #270 mesh sieve. Another common way to classify sands is material passing a #4, or 6 mm (0.25") sieve.

Sands help fill in voids between larger aggregates.

Gravel – Gravel is also produced by the natural disintegration of rock and is found around the world. Gravel is larger than sand, the dividing point being the #4 mesh sieve or 6 mm (0.25"). Gravel typically has rounded surfaces. To be included in a bituminous mixture, gravel is usually crushed, not just for sizing purposes, but also to create angular faces that are more desirable for building strength and stability in the bituminous layer.



There are two types of gravel: bank and river. Bank gravel is mined from deposits in areas defined by glacial activity. Bank gravel can be clay-like or sandy, depending on the proportion of fine to coarse material. Bank gravel must be thoroughly washed prior to stockpiling for use at an asphalt manufacturing facility.

River gravel is composed of small pieces of rounded stone, usually no larger than a large coin, harvested from a river bed. The stones are typically more rounded due to erosion caused by water flowing over them for many years.

Crushed Stone – Crushed stone results from the mechanical reduction of gravel, rock, and boulders by specially designed crushing equipment. During the crushing operation, oversized chunks are removed by a scalping screen. A series of vibrating screens can be used to further grade the crushed material.

Crushed sedimentary rock, formed from sediment in water or from wind-blown deposits is one type of aggregate. Sandstone, made up of mineral fragments compressed into layers over a long period of time, is an example of sedimentary rock. Limestone is a sedimentary rock made up of fossilized and compressed plants and animals. Gypsum is a sedimentary, layered rock that results from evaporation or a chemical process.

Sedimentary rocks are plentiful in some areas and are relatively easy to mine and crush. Some sedimentary rocks are fragile and can be damaged during the paving and compaction process.

Igneous rocks, a second class of crushed aggregates, are the result of cooling and hardening of hot molten materials originating beneath the surface of the earth.

Extrusive igneous rocks, like basalt, are those that cooled on the surface of the earth. Intrusive igneous rocks cooled and formed below the surface and were later exposed by erosion or movement of the earth's surface.



Consistent aggregate size and type are keys to setting up a paver to achieve best results.

Igneous rocks are harder than sedimentary rocks and more durable. The drawback to the use of crushed igneous rock is accelerated wear on paver and screed components due to the hardness of the aggregate.

Metamorphic rocks were formed from sedimentary and igneous rocks by prolonged exposure to pressure, high heat and/or chemicals. Metamorphic rocks like gneisses, schists and quartzites are hard, but brittle. They can be damaged by excessive impacts during compaction or excessive tamping force.

Slag – There are various types of slag, a nonmetallic byproduct of a blast furnace or coal combustion in a wet-bottom boiler. Quality requirements for the use of slag in bituminous mixtures vary considerably, but slag can be a good paving grade aggregate when properly sized.

Many slags are porous and absorb bitumen (asphalt cement) readily. Bitumen content is normally adjusted upward when slag is used.



Carefully metered aggregates, bitumen and filler materials produce consistent asphalt quality

Mineral Fillers – Mineral fillers, often referred to as fine materials, are defined as any materials that will pass a #200 mesh screen. The minerals fill in the very small voids in the bituminous layer and thus increase the stability of the layer.

Mineral fillers combine with the bitumen in some mix designs to create a mastic that helps bond the aggregates. Mineral fillers can be crusher dust, Portland cement, flyash, hydrated lime, or naturally occurring fine deposits such as loess. Mineral fillers have special storage containers at the asphalt manufacturing facility and are used in carefully metered amounts.



[TYPES OF BITUMINOUS MIXES]

Bituminous mixes are classified in two broad categories: hot mix asphalt (high-temperature asphalt) and warm mix asphalt (low-temperature asphalt).

Hot mix asphalt – Hot mix asphalt, or high-temperature asphalt, is the product originally produced by modern asphalt manufacturing facilities. In order to lower the viscosity of the straight-run or modified bitumen, the storage temperature of the bitumen will be in the range of 150° C to 175° C (300° F to 350° F).

The aggregates will be heated to the same temperatures prior to blending and coating with the bitumen. These elevated temperatures are necessary to assure proper coating of the aggregates by bitumen and to aid in the compaction process. Depending on the mixer output temperature, ambient conditions and length of haul, the bituminous material may pass under the screed at 150° C (300° F), for example.

Warm mix asphalt –The manufacture and installation of warm mix asphalt, or low-temperature asphalt, started during the 1990s in Europe. The goal of the early research was to find methods to manufacture bituminous material at significantly lower temperatures (using less fuel and creating fewer emissions) while maintaining workability, compaction and serviceability that are equal to or better than hot mix asphalt.

Warm mix asphalt is now accepted as a viable alternative to hot mix asphalt around the world, and the range of applications has grown to nearly universal acceptance.



The texture of the mat is influenced by the type and size of the aggregate.

Warm mix asphalts are produced by adding one of three additives to straight-run bitumens:

1. Water foamed asphalt with an application temperature of 129° C to 135° C (265° F to 275° F)
2. Organics or waxes.
3. Chemicals or surfactants. For wax or chemical additives, the temperature range is 116° C to 121° C (240° F to 250° F).

Dense-graded mix – Dense-graded mixes are produced with continuously graded aggregates. In other words, a variety of aggregate sizes are utilized in the design, along with fines and bitumen. Dense-graded bituminous material is used for many paving applications and may be installed as base layers, binder layers or surface layers. Dense-graded mixes can be further classed as coarse (harsh) mixes or fine (tender) mixes.

Coarse mixes have maximum nominal aggregate size of 19 mm (0.75") or larger in their designs. Coarse mixes tend to support the screed better than fine mixes, but coarse mixes are more prone to segregation.

User Tip: *When laying down a coarse mix, pay special attention to end-of-load segregation and segregation due to erratic feeder system operation. Minimize or discontinue cycling hopper wings to prevent segregation. Do not pave out the surge capacity in hopper inserts or material transfer vehicles when paving with large stone mixes. Adjust the feeder system to run continuously with auger speed in the range of 20 to 40 revolutions per minute.*

Fine mixes have maximum nominal aggregate size of 12 mm (0.5") or less and often a higher percentage of bitumen. Fine mixes are more easily deformed by the weight of the screed, but are usually less prone to segregation.

User Tip: *When laying down a fine mix, you may have to engage the screed counterbalance system to help the screed float correctly and to avoid deforming the asphalt layer during paver stops. If the paver is so equipped, engage the screed support system during paver stops to prevent deep marks.*

Open-graded mix – Open-graded mixes have relatively uniform-sized aggregate typified by an absence of intermediate-sized aggregates. Open-graded mixes are most often used as surface layers, but may also be used as permeable base layers.

Open-graded surface layers utilize small aggregates (usually less than 12 mm / 0.5") and are laid down in thin layers. Because thin layers of bituminous material can lose heat quickly, it is very important for the crew to maintain a speed that allows the compaction train to keep pace with the paver. Mixes like these typically do not have segregation issues. But, because there is a lot of stone-on-stone contact within the layer, watch for aggregate damage when using tamping screeds.



Layers of open-graded mixes lose heat rapidly so paving speed must be matched to the speed of the breakdown roller.

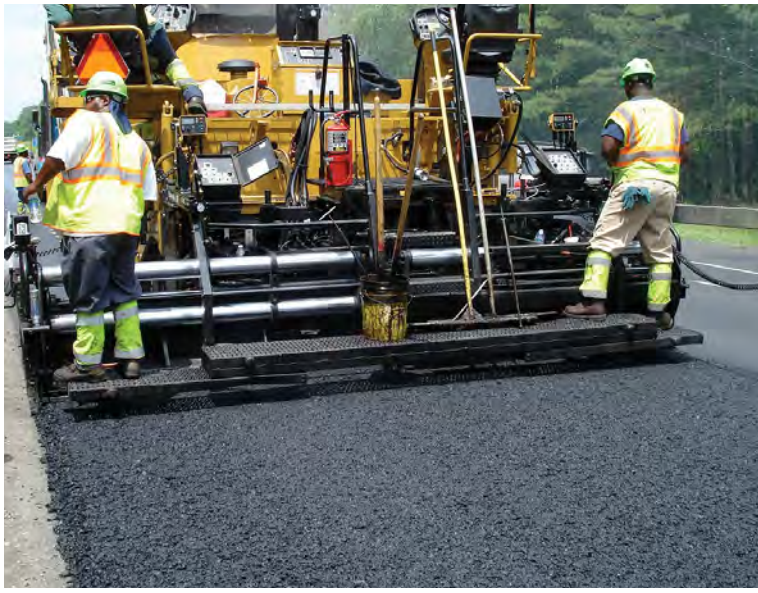
Gap-graded mix – Gap-graded mixes use an aggregate gradation with particles ranging from coarse to fine with intermediate sizes missing or present in small amounts. These mixes also have a high degree of stone-on-stone contact, meaning the same precautions must be observed as when paving with tamping screeds to avoid layer damage. Gap-graded mixes are also more permeable than dense-graded mixes.

Stone-matrix asphalt – Stone-matrix asphalt, like other open-graded mixes, will be missing intermediate aggregate sizes. Stone-matrix asphalt consists primarily of large aggregates and fines and is bound by modified bitumen. These types of mixes are very stiff because of the modified binder and are difficult to move by hand after passing under the screed.

Stone-matrix asphalt is usually very temperature sensitive and must be compacted at elevated temperatures. Therefore, this is another mix that requires the paving crew and the compaction crew to work at the same pace and keep the rollers in the same temperature zone.

Porous asphalt – Porous asphalt is a special mix design created to allow water to percolate through the bituminous layer and underlying aggregate base layer so the runoff water can be captured in a storage reservoir.

Porous asphalt has no fine material in its gradation. Air void content varies considerably according to the engineering requirements. Air voids may be as low as 15 percent or as high as 28 percent in other areas. Porous asphalt is a surface layer, usually laid down between 5-10 cm (2-4") thick. Porous asphalt is laid down using all types of screeds, with or without screed vibration and tamping forces.



Stone-matrix asphalt is a stiff mix that is difficult to rake and shovel.

[BITUMINOUS LAYERS]

Depending on the type of structure that is being built or maintained, as many as five consecutive bituminous layers over a base or sub-base may be placed. Or, may be placed just one layer of bituminous material on some sort of base material. Each layer has its own purpose, usually some sort of mix design unique to that particular layer, and a thickness that satisfies the requirements of that layer.

Let's examine the layers that make up a highly-engineered highway project. Many such projects use a design known as perpetual pavement.

Bituminous base layer – The base layer of a highway project is laid down on a compacted granular base or stabilized sub-grade and base. In a perpetual pavement design, the base layer may be up to 30 cm (12") thick. The role of the base layer on a highway project is to resist the strain caused by heavy loading and prevent deformation of the underlying base and/or sub-grade.

The aggregates in the bituminous base layer are usually large—up to 38 mm (1.5"). It is common to pave these layers up to 15 cm (6") thick. The base layers are also the first step in creating a smooth longitudinal profile. Averaging skis are commonly used when paving base layers.

Bituminous binder layer – The primary role of the binder, or intermediate, layer of a highway project is to add strength. In this case, the strength is meant to prevent rutting that may occur over time in the surface layer from extending into the intermediate layer.

The mix used in the binder layer will usually have aggregate slightly smaller than the base layer, but is normally a high-quality, dense-graded mix, often using modified bitumen for stiffness. Averaging skis are frequently used when paving binder layers to make additional gains in smoothness. Binder layers often include about 15 cm (6") of bituminous material.

Bituminous surface layer – The surface layer, normally around 5 cm (2"), may have several roles. It may be an impermeable mix designed to keep moisture away from the layers below. Or, as previously discussed, the surface layer may be open-graded to allow moisture to percolate down and prevent puddles and hydroplaning.

The surface is usually the stiffest mix to help minimize rutting caused by heavy axle loading. It should have a skid-resistant texture and the correct transverse profile for drainage. The surface layer is normally the thinnest layer.

A bituminous layer may serve many functions, especially on lower-volume streets and roads. On many streets and parking lots, only two bituminous layers are used to provide strength. On some rural roads, only one bituminous layer is placed on a granular base. In these instances, the granular base or sub-grade provides the bulk of the structure's load-bearing capability. The bituminous layer(s) prevent moisture penetration and provide a durable, smooth surface.



Leveling layer – On some projects, especially cold plane and re-pave projects, a leveling layer, also called a shim layer, may be installed before the surface layer. The purpose of the leveling layer is to correct the transverse profile (slope) and to help improve the longitudinal profile (smoothness). A leveling layer is usually 5 cm (2") thick at the most.

User Tip: *If slope correction is required while paving a leveling layer, check the slope of the base prior to paving. Knowing the amount of slope corrections that will be required will help determine whether to pave slopes using automatic slope control or manual slope control. In Unit 6, the use of slope is discussed in greater detail.*

[SUMMARY]

Paving crewmembers need to know enough about mix designs and the production of bituminous material to help them plan the paving process and to troubleshoot problems on the job. Above all, the supervisor of the paving crew must stay in communication with the asphalt manufacturing facility. Any changes in hourly production will require changes to paving speed and compaction patterns.

Even minor changes in mix design, binder content and gradations can have a large impact on the quality of the bituminous layer and the compaction process.


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Unit 2
PRE-PROJECT PLANNING

Never underestimate the importance of pre-project planning. Plan early, and create a process to ensure all elements of the plan are in place and executed.



Many common quality problems disappear when proper planning is done before paving begins, and when good communications are in place with all those involved in the process.

The stages of pre-project planning vary, depending on the size and scope of the project. In some instances, planning includes a formal document that is written by the paving contractor and submitted to the project owner for approval.

In other cases, especially on small projects, planning is less formal. But in any situation, planning should not be left to the last minute. A pre-project checklist that is reviewed and approved before the crew arrives at the project should always be used.

A discussion at the start of the shift is a good idea but should not be substituted for thorough planning days or weeks before the project starts.

In this unit, a comprehensive checklist will be developed. Although some of the usual steps are missing, that is okay. Keep the steps in place on the project checklist. Likewise, not all the items mentioned in this unit will apply to every project.

Typically, a project manager is the primary planning coordinator. Depending on the size and structure of the organization, the role of the project manager may be delegated to several people, such as an owner, a superintendent, a crew supervisor or a quality control manager.

[ROLE OF THE PROJECT MANAGER]

The project manager is the link between the project owner, the engineering firm (if any), the general contractor (if any), subcontractors, suppliers and the various departments within the asphalt paving company. The project manager helps to create the plan and makes sure that all steps are followed by the asphalt manufacturing facility, the quality control team, subcontractors and the paving and compaction crewmembers.

[PRE-CONSTRUCTION MEETING]

Most large projects have formal pre-construction meetings that help to finalize the construction plan. The project manager is the primary representative for the paving contractor and may be assisted by a quality control manager and project superintendent. During the pre-construction meeting, many items in the plan are discussed, reviewed and agreed upon. Here is a typical list of points to be covered.

1. Describe the scope of the project.
2. Discuss project specifications to clarify any questionable aspects before starting construction.
3. List project personnel and establish clear lines of authority, as well as contact information.
4. Discuss sequencing, scheduling and methods for completing a quality project on schedule and within budget.
5. Provide a paving plan to include such items as stockpiling of material; plant production and operation; scale operations and procedures; haul units and routes; paving widths; speed of paving; and type and operation of equipment to be used.
6. Provide a plan for constructing a control strip (test section) of adequate size to simulate actual production paving rates and conditions.
7. Discuss sampling and testing frequency and locations, re-testing procedures and quality control and acceptance of program details.
8. Discuss safety goals and procedures to be used on the project: for example, weekly safety meetings with all project personnel.
9. Set up a means to have on-going communication with the owner's representatives and project representatives. Arrange weekly meetings to focus on schedule updates, personnel changes and use of different equipment or methods. Also discuss problems that have occurred or might occur.
10. Discuss traffic control and the best ways to keep the public aware of project progress and planned activities. Topics should include use of media, message boards, flyers to local businesses, local council meetings and more.

[SPECIFICATIONS AND REQUIREMENTS]

Method specifications – The public works agency, especially on highway projects, may write method specifications that determine what type(s) of equipment and construction techniques must be used, as well as any other special conditions that must be addressed. Here are some examples of typical method specifications.



Use of a material transfer vehicle is frequently written into the specifications on highway projects in North America.

The use of material transfer vehicles and their functional capabilities is specified on some highway projects. Those capabilities may include a minimum surge capacity and a type of remixing or blending system. The intent of this specification is to promote continuous paving by separating the paver from the haul units. (See Unit 7 for more information about material transfer vehicles.)

Sometimes the type and size of compaction equipment is controlled by a written method specification that varies depending on the type of bituminous material, layer thickness, and layer purpose.

To promote the construction of durable longitudinal joints, some highway owners require staggered or echelon paving. Staggered paving requires two or more pavers in a style that creates a hot-on-hot longitudinal joint. Staggered paving requires careful planning for haul unit operation, material transfer vehicle usage, and paving speeds. (See Unit 7 for more information about echelon paving.)

An example of a mandatory special condition is the requirement to use attachments that create a wedge joint or notched wedge joint at the unconfined edge of a paving line. These specialty joints require installation of joint formers at the outer end of the screed.

In each of the previous examples, the project manager has to communicate the needs to the equipment manager or equipment dispatcher to assure that the right equipment is available for the project.

End result specifications – End-result specifications typically describe some sort of measurement or testing to verify the quality of the finished product. End-result specifications often dictate how the crew will set up and operate the equipment and, therefore, must be understood by the crew before they begin working.



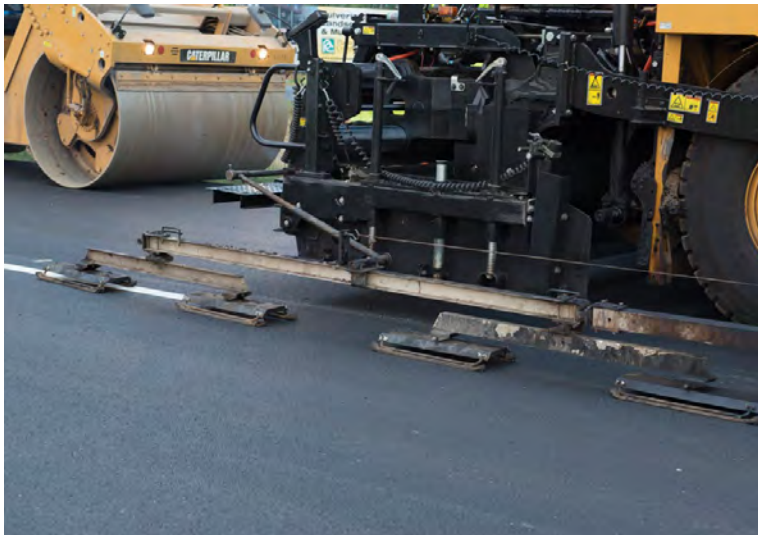
A notched wedge joint is mandatory on this project.

Highway projects may have a performance pay factor, or some acceptance factor, based on the smoothness of the bituminous layer. Common ride quality measurement systems used today are the International Roughness Index, Profile Index, and Ride Number.

In each case, a target will be set for ride quality. Depending on the public works department's policy, the contractor may earn an adjustment in pay per ton (either upward or downward), or may have to perform corrective action to reach a minimum target for ride quality. On projects with a ride-quality specification, use of averaging skis is common. (See Unit 6 for more information about paving with averaging skis.)

Most projects have some sort of end-result specification for in-place density. Depending on the type of mix and the difficulty of the specification, the project manager may have to order additional compaction equipment or specialized compaction equipment. An airport project, for example, requires more planning than a street project because there typically is a higher risk factor associated with creating the specified densities.

Another requirement that is written with increasing frequency measures the uniformity of the temperature of the bituminous layer behind the paver and prior to compaction. Quality control



Mechanical averaging skis are often used to help meet ride quality specifications.

personnel have to know how to document the temperatures to satisfy the specification. The paving crew has to know how to control the temperature of the layer and how to take corrective action when temperatures vary too much. In this example, the project manager will probably order the use of a material transfer vehicle and set a paving speed to establish non-stop paving.

Requirements – Job requirements may not be written specifications. The requirements can simply be the most important results to the owner. If the crew talks about and understands these requirements, they can usually set up and operate the paver to achieve the results.

In a parking lot, the most important concerns are usually drainage and appearance. Before paving, the crew should check the slope of the base to determine if the drainage created by the base prep crew is correct. If the slopes are correct, the paver can be set up to lay down the specified thickness and water will drain appropriately.

If the base slopes are incorrect, the crew may decide to set up the paver using automatic slope control. If the slopes are significantly wrong, the crew that installed the base should be brought back to correct the defects.

User Tip: *Remember, when slope control is used, the layer thickness will vary and yields can be adversely affected. Always try to correct slopes prior to paving. Use slope control to make minor slope corrections. (See Unit 6 for more information about slope control.)*

To improve the appearance of a parking lot surface, the crew should lay out the project to maximize the creation of hot-on-hot longitudinal joints or warm-on-hot joints. The compaction pattern is then planned to overlap the joints immediately for best appearance.

In another application, height match to an existing structure such as a concrete gutter may be the most important requirement. In that situation, the crew will need a grade control sensor and the necessary mounting equipment. Prior to paving, check the condition of the gutter to verify that it is suitable to use as a grade reference.



Drainage and appearance are important to the owners of parking lots.



On airport projects, elevations and yield are often the most critical requirements.

On some projects, yield or layer thickness, is the priority to the owner of the bituminous structure. The crew needs to know beforehand when a project has been bid as a "yield" project. They will set up the paver to maintain a uniform layer thickness and typically will not be using averaging skis to control yield.

In summary, specifications and requirements have to be understood by managers, crew supervisors, quality control staff and crewmembers.

[MOBILIZATION AREAS]

The equipment that is going to be used has to be delivered to the project, unloaded and parked at a pre-arranged location. Normally, these areas are referred to as mobilization, or staging areas.

The project manager may have to rent space to park equipment between shifts, if the work will continue for an extended time. The manager will have to find several parking locations if it's a highway project that covers a significant length in terms of kilometers or miles paved.

The mobilization area may include space for crew parking, or the project manager may have to organize transportation for crewmembers. The mobilization area may include space for equipment cleaning. In that case, the crew must be aware of local regulations that control the placement, clean-up and salvage of excess bituminous material. The equipment manager and dispatcher must be informed about the locations of all mobilization sites.

[TRAFFIC CONTROL]

Traffic control affects the safety of the motorists and the workers on the project. An approved traffic control plan should be in place. The plan must conform to local regulations concerning lane closures, detours and time available for work on any given day.

Following is a list of questions that may need to be answered. There may also be special considerations in the local area. Always check with local law enforcement and/or the public works department to resolve any questions.

1. Can newspapers, radio or television be used to give the public advance notice of delays and detours?
2. Can message boards be used to alert motorists in advance of the project? Do the motorists have enough time to react to instructions?
3. Can motorists understand what they need to do?
4. Are a speed detecting radar device and message board available?
5. Are road closures possible, instead of lane closures, to keep traffic away from the work zone?
6. Are adequate barriers, cones and barrels in place to clearly guide traffic away from work areas?
7. Do traffic control personnel have the proper communication devices?
8. Are local law enforcement officials available to assist with traffic control?
9. Are provisions made for changing lane closures?
10. Has someone planned for temporary striping?
11. Is a plan in place for maintenance of warning devices such as message boards and flashing lights?



Lane closures on major highways require planning and good communication for the motorists and crewmembers.

12. Are intersections properly controlled?
13. Are provisions in place for helping haul units enter and depart the work zone?
14. Are provisions made to relieve flaggers and drivers of pilot vehicles?
15. Are all personnel equipped with reflective personal protective gear?
16. Is the proper lighting arranged for night projects?

A great deal of the work for traffic control may be done by a subcontractor, but ultimately the project manager is responsible for ensuring that the traffic control plan is understood by all and continuously monitored.

[HAUL UNIT CLEANING SITE]

Before returning to the asphalt plant, haul units must be cleaned to prevent spilling bituminous material on roadways. Whenever project layout permits, locations should be designated where haul units can be cleaned. These locations should be indicated on the approved haul unit route and given to each driver.

There are several reasons why clean-out areas should be used. First, if haul units are cleaned out directly in front of the paver, the piles of material will lose heat and will show up as cold spots in the paved layer when the screed passes over or through the piles.

Second, the piles may be run over by the tires of the next haul unit, thus creating high spots in the grade being paved. The screed may then drag over the high spots, creating blemishes and areas of open texture.

Finally, requiring a crew to clean spills will stop the paver and that may contribute to deformation and inconsistent temperature of the layer.

The best practice is to have haul units clean out at a designated area and then salvage that material at the end of the shift.



On this project, drivers use the emergency lane as a designated area for vehicle cleaning. Material will be salvaged at the end of the shift.

[GRADE CONDITIONS]

It is extremely important that grade conditions be checked and corrected prior to paving. No matter what type of grade is in front of the paver—crushed aggregate base, reclaimed bituminous base, milled surface, existing bituminous surface, or concrete—defects in the grade can cause defects in the bituminous layer.



Substandard base, like this weak area, must be identified and repaired prior to paving.

Aggregate or reclaimed bituminous base – The preparation of the base may be done by a subcontractor or by a division of the paving contractor. The base crew has the responsibility to meet the specifications, and stringent quality control should be in place. Prior to paving, here are some items to verify.

1. Has the base been inspected and accepted? Who signed off on the quality of the base? Is the inspector available for consultation if any issues need to be discussed?
2. Has the base been proof rolled? Is a proof roller (usually a pneumatic compactor) still available on the project?
3. Is a motor grader available to make corrections if needed? Is a compactor available to work behind the motor grader?
4. Is a water truck available for dust control and/or to add water for additional compaction?
5. Has the slope of the grade been checked in numerous locations to verify that the transverse profile conforms to plan drawings? If the paver's slope control is needed to correct slopes, has the project inspector been notified that yields may be adversely affected?
6. Are super-elevations, if any, clearly marked so screed operators can see where transitions are required?
7. Is a prime coat required on the aggregate base? Is the application rate correct? How much time is needed for the prime coat to cure?

Milled surface – Milled surfaces require inspection before paving. The milling contractor often performs the inspection, and it is the project manager's responsibility to ensure that step has been taken. Paving should not proceed until the quality of milled surfaces has been verified.

1. Are the elevations left by the cold planer correct? In other words, has the correct amount of material been removed? This is important when there are height clearance issues on the project or when there is a need to match the height of adjacent structures such as curbs and gutters.
2. Is the transverse profile (slope) correct? Remember, a cold planer can be set up to control slope just like a paver.
3. Is the surface texture acceptable? There are different types of cutter drums. A project may require micro-milling produced by a cutter with closely spaced tools. Also, excessive milling speed can cause the tool spacing pattern to be incorrect.
4. Has the cold planer created a rough ride or has the cold planer improved ride quality? It is a good idea to verify the ride quality of the milled surface when the cold planer starts to work on a project. If there is a high degree of roughness, the operation of the cold planer should be changed (install averaging ski, slow down, increase tool maintenance, for example) to improve the smoothness of the milled surface.



If the milled surface has areas of delamination, the milling depth may need adjustment.

5. Are there areas of delamination? When the depth of the cut is just above the intersection of two bituminous layers, a portion of the upper layer may lose its bond with the lower layer and break off in thin chunks. Paving over delaminated areas creates variable thickness with resulting loss of smoothness and variations in density. The project manager may ask for a change in the planned cutting depth to reduce delamination.
6. Has the surface been properly swept, flushed and cleared of all debris? Paving should not begin until all loose material is removed from the milled surface.

Overlay – Paving over existing pavement layers has many of the same issues as paving over a milled surface. For example, the crew should verify the slopes and have a good idea of what needs to be done to improve smoothness. Following are other considerations:

1. Have all badly deteriorated areas, such as block cracking, been corrected?
2. Have all potholes been patched or filled?
3. Have all high and/or low spots been removed or pre-leveled?
4. Have cracks been sealed or filled? Some types of crack sealant tend to expand once hot bituminous material has been placed over them. Be careful not to use too much crack sealant, and to use one that does not react to hot mix.
5. Have all existing joints and tie-ins been cut to create a straight vertical face?
6. Has the surface been swept and cleaned?
7. Is the tack distributor properly calibrated to apply the correct amount of tack?
8. How much curing time is required before paving over the tack being applied?
9. Has an approved release agent been applied to structures such as manhole covers and water valves?

[PROJECT LAYOUT]

Prior to paving, analyze the project to determine the most efficient way to complete the job. Some projects are straightforward in their design and only need a brief review with crew supervisors and crewmembers. Other projects, like large parking lots, must be carefully planned and laid out. Following are some items to consider when laying out the project.

1. Lay out the project with the fewest number of passes possible to minimize the number of longitudinal joints.
2. Designate crew and allow time for the crew to paint paving width guides or to install string to determine paving widths.
3. Check the specifications to determine how to stagger longitudinal joints on multiple lift projects.
4. If the screed will not reach the widest planned width, be sure to order bolt-on screed extensions or, if a larger paver is available, order the larger paver.

User Tip: *Most pavers can be ordered with wide-width paving kits. The kits contain bolt-on screed extensions, auger extensions, auger support systems, and mainframe (tunnel) extensions. Always install the appropriate auger and mainframe extensions to facilitate the movement of material to the outer edges of the paving width. (See Unit 3 for more information.)*



Order the necessary extensions for screeds, augers and mainframes when preparing for wide-width paving.

1. Plan for constant paving widths whenever possible. It is especially important to avoid changing width while paving on projects with ride quality specifications. For most commercial projects, moving hydraulic screed extenders in and out is unavoidable and is usually acceptable.
2. Pave emergency lanes (shoulders) at the same time as driving lanes when possible. If the emergency lane has a different slope, use the screed extender to create the desired slope.
3. If the emergency lane must be paved separately, pave it first whenever possible. The emergency lane will serve as a grade reference for the adjacent traffic lane, and also can serve as a place to stop and reverse compactors, and to park compactors when refilling water spray systems.
4. Order cutoff shoes when the paving width is less than the basic width of the main screed. An example would be when using a 3 m (10') paver to pave a 2.5 m (8') emergency lane.
5. Plan the setbacks for each shift. On some multi-lane projects, exposed longitudinal joints cannot be left. That requires backing up and matching each lane that is paved during the shift. The setbacks take time and should be included in calculations for paving speed and daily production.



Paving the emergency lane first provides a better reference for the grade control system when paving the adjacent driving lane.

6. Pave the primary driving lanes prior to paving intersection approaches, tie-ins, turning lanes, tapers and ramps. This is especially important when the project has ride quality specifications. Designate a second crew to pave the small areas and do not interrupt the high-production work of the mainline crew.
7. Select grade controls that match the project layout. For example, if paving in one lane in one direction (no setbacks) for the entire shift, a mechanical ski will likely work best. Be sure to have the correct ski installation hardware. Or, the project may include obstacles such as catch basins that prevent use of a mechanical ski. In that case, a non-contact ski that requires different mounting hardware will likely be selected.

User Tip: *Be careful not to create a situation where haul units must back up a long distance to reach the paver that is doing setback paving. This can cause a steep drop in production. Avoiding this situation is usually easier on highway projects than parking lots and city streets.*

[ORDERING HAUL UNITS]

Ordering the correct number of haul units has a major affect on hourly production and the working speed of the paver. The number of haul units should be sufficient to permit continuous, maximum operation of the asphalt manufacturing facility and, thus, maximum hourly production. The hourly plant production is one of the inputs needed to calculate the working speed of the paver.

The project manager usually delegates ordering haul units to the project superintendent or to the crew supervisor. Calculate how many haul units are needed by gathering the following information:

Planned hourly output of the asphalt manufacturing facility. The maximum production rate of the asphalt plant is a known factor, but remember the production rate can be affected by moisture content in the aggregates and other variables.

Plan amount of material held in storage silos at the start of the shift. For example, if the plant can produce 200 tonnes (220 tons) per hour, and the plant is going to start production an hour before haul units are scheduled, then 200 tonnes (220 tons) will be stored.

Cycle time for a haul unit to make a round trip from the plant to the job and back. The cycle times may vary depending on traffic volumes, especially if the haul route is in an urban area. The number of haul units may have to be increased or decreased to account for peak-hour traffic conditions.

Average haul unit load. If all the haul units have basically the same capacity, the average load is easy to determine. If the capacities are different, some extra calculations will be required to find the average load carried by the haul units and determine the appropriate arrival sequence at the worksite.



Having haul units with different capacities scheduled for a project complicates ordering the correct number of haul units.

Example One, No Storage – For this project, the asphalt manufacturing facility is going to produce 200 tonnes (220 tons) per hour. There are no storage bins. The cycle time for haul units is 90 minutes (1.5 hours). The average load for the haul units is 24 tonnes (26.5 tons).

200 tonnes per hour ÷ 24 tonnes =
8.33 trucks per hour

8.33 trucks per hour x 1.5 hours =
12.5 trucks

In other words, 13 trucks would be needed to have adequate capacity to match plant output under these conditions.

To calculate the number of haul units:

1. Tons per hour production ÷ average load = trucks per hour
2. Trucks per hour x cycle time in hours = total trucks needed

Example Two, Storage Bins – Next, add storage bins to the calculation. Assume 250 tonnes (275 tons) in storage at the start of the shift. Assume that the crew is scheduled to work for 10 hours during this shift. Make the calculation to use a portion of the storage each hour so the combined hourly production with stored material will be used uniformly by the end of the shift.

250 tonnes ÷ 10 hours =
25 tons per hour added production

Therefore, hourly production is increased to 225 tonnes (250 tons) per hour. Calculate the haul unit requirement again using the new production figure.

225 tonnes per hour ÷ 24 tonnes =
9.38 trucks

9.38 trucks x 1.5 hours = 14 trucks

To transport the additional hourly production, order at least one more haul unit.

Example Three, Variable Capacity – Assume that two different types of haul units are available for the project described in Example Two. The paving company owns haul units that carry 24 tonnes (26.5 tons). There will be eight of those haul units assigned to the project. Other haul units will be hired, but they transport an average of 20 tonnes (22 tons). How many of the smaller haul units are needed?

First, calculate how many tonnes the eight larger trucks can transport when they all complete a round trip and then divide by the cycle time.

8 trucks x 24 tonnes ÷ 1.5 hours =
128 tonnes/hour

Next, subtract the weight carried by the larger trucks from the total hourly production.

225 tonnes minus 128 tonnes =
97 tonnes per hour remaining

Finally, revert to the original formula.

97 tonnes per hour ÷ 20 tonnes =
4.85 trucks

4.85 trucks x 1.5 hours = 7.3 trucks

Therefore, eight of the smaller haul units would be hired to complement the eight larger company owned units.

Example Four, Variable Traffic – In the previous examples, the cycle time—1.5 hours—has been constant. Now, assume that the haul units carrying 24 tonnes (26.5 tons) will be traveling through peak traffic for two hours at the start of the shift and for two hours at the end of the shift. During peak traffic hours, the cycle time will increase to 2.2 hours. Use the formula to determine the correct number of haul units.

For the period 0700 to 0900 21 trucks would be hired. Between 0900 and 1500, seven trucks would be assigned to hauling aggregates or working with a cold planer, for example. Then, direct the seven trucks to return to the paving project to work from 1500 to 1700.

The task of making these calculations is simpler using software like the Paving Production Calculator. This software is available as an App from the appropriate App stores. Shown below are the previous examples as calculated using the Paving Production Calculator.

225 tonnes per hour ÷ 24 tonnes = 9.38 trucks

9.38 trucks x 2.2 hours = 21 trucks

During off-peak traffic hours, the cycle time reverts to 1.5 hours.

225 tonnes per hour ÷ 24 tonnes = 9.38 trucks

9.38 trucks x 1.5 hours = 14 trucks

Example 1, No storage — For this project, the asphalt manufacturing facility is going to produce 200 tonnes (220 tons) per hour. There are no storage bins. The cycle time for haul units is 90 minutes (1.5 hours). The average load for the haul units is 24 tonnes (26.5 tons).

TRUCKING CALCULATOR

Trucking	**General Inputs**		*Imperial Units*	*Metric Units*
Paver Speed	Production Rate of Hot Plant:		[220] *tons/hr*	[200] *tonnes/hr*
Compaction	Multiple Silo Plants: Initial Storage		[0] *tons*	[0] *tonnes*
Windrow	Paving Hours:		[0.0] *hrs*	[0.0] *hrs*
Yield	Truck Capacity (size):		[26.5] *net tons*	[24.0] *net tonnes*
Slope	**Truck Cycle Times** *(minutes)*			
Thickness	Load Time and Ticket:	[6]		
Job Summary	Tarp:	[4]		
Legal	Haul to Job:	[35]		
	Time on Site:	[2]		
	Dump / Clean:	[8]		
	Return Haul:	[35]		
Exit	**Truck Cycle Factor** *(total time in hours)*	[1.5]		
	Number of Trucks Needed:	[12.5]		

Example Two, Storage Bins – Next, add storage bins to the calculation. Assume 250 tonnes (275 tons) in storage at the start of the shift. Assume that the crew is scheduled to work for 10 hours during this shift. The calculation is made to use a portion of the storage each hour, so hourly production will be combined with stored material to use the material uniformly by the end of the shift.

250 tonnes ÷ 10 hours =
25 tonnes per hour added production

Therefore, hourly production is increased to 225 tonnes (250 tons) per hour. Calculate the haul unit requirement again using the new production figure.

TRUCKING CALCULATOR

Trucking
Paver Speed
Compaction
Windrow
Yield
Slope
Thickness
Job Summary
Legal
Exit

General Inputs	*Imperial Units*	*Metric Units*
Production Rate of Hot Plant:	[220] *tons/hr*	[200] *tonnes/hr*
Multiple Silo Plants: Initial Storage	[276] *tons*	[250] *tonnes*
Paving Hours:	[10.0] *hrs*	[10.0] *hrs*
Truck Capacity (size):	[26.5] *net tons*	[24.0] *net tonnes*

Truck Cycle Times *(minutes)*	
Load Time and Ticket:	[6]
Tarp:	[4]
Haul to Job:	[35]
Time on Site:	[2]
Dump / Clean:	[8]
Return Haul:	[35]

Truck Cycle Factor *(total time in hours)*	[1.5]
Number of Trucks Needed:	[14.0]